UCT41042 – Practical for Introduction to Smart System

Department of Information & Communication Technology

Faculty of Technology

South Eastern University of Sri Lanka

Supportive Document for VS code

Title: Create and Run ESP32 Project using Wokwi Simulator and platformIO extension in Visual Studio Code.

Part A: Setup Wokwi in Visual Studio Code

1. Install Required Extensions in VS Code
    - Open Visual Studio Code.
    - Go to Extensions.
    - Install the following:
        - Wokwi Simulator
        - PlatformIO

2. Activate Wokwi License
    - Press F1 to open Command Palette.
    - Type and select: Wokwi: Request a new License
    - Click Open when prompted.
    - It will open your browser. Log in to your Wokwi account.
    - Click GET YOUR LICENSE.
    - Return to VS Code and confirm the license activation.

Part B: Create a New ESP32 Project in VS Code

1. Open the PlatformIO sidebar (ant icon on the left).
2. Click on New Project.
3. Fill in the details:
    - Name: ESP32-FirstProject
    - Board: Espressif ESP32 Dev Module
    - Framework: Arduino
    - Choose your desired location → Click Finish.
4. Wait until PlatformIO finishes initializing the project.

Part C: Add Wokwi Simulation Support

After the project is created, you need to add two important files for Wokwi:

1. In the root of your project, create a new file named wokwi.toml and paste the following:

    ```
    [wokwi]
    version = 1
    firmware = ".pio/build/esp32dev/firmware.bin"
    elf = ".pio/build/esp32dev/firmware.elf"
    ```
2. Create a diagram.json file in the root to define hardware.
3. Right click diagram.json file and select open with, then select Text Editor.
4. Copy and paste here, the code from digram.json file in Wokwi site.

Part D: Add the code for ESP32

1. Open the file:
    - src/main.cpp
2. Remove default codes except #include <Arduino.h>
3. Copy and paste here, the code from sketch.ino file in Wokwi site.
4. Save the file.

Part E: Build and Run the Simulation

1. Press f1 → Click on PlatformIO:Build → (or press the ✓ icon at the bottom).
2. After successful build, press F1 → type and select: Wokwi: Start Simulator

How to Add Libraries for ESP32 in Visual Studio Code using PlatformIO

Method: Using PlatformIO Library Manager

1. In the PlatformIO sidebar, click on PIO Home → Open.
2. In the PlatformIO Home page, click on the Libraries tab (on the left menu).
3. In the search bar, type the name of the library you want to add.

   Example: Type WiFiManager

4. Find the correct library in the results (for WiFiManager, choose tzapu/WiFiManager).
5. Click the Add to Project button.
6. Select your current project from the list and click Add.
7. PlatformIO will automatically add the library to your platformio.ini file.
8. After adding the library, you can use it in your code like this:

```
#include <WiFiManager.h>
WiFiManager wm;
```